

CS 6371 - Assignment 9

Christian Parker
CXP220034@utdallas.edu

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1)

$$\frac{\frac{D_1 \quad D_2}{\{x = \bar{i}\} \quad c_3 \quad \{C\}} \quad (3) \quad \frac{}{\{C\} \quad y := y + x \quad \{y = \max(0, 2\bar{i})\}} \quad (4)}{\{x = \bar{i}\} \quad c_3; y := y + x \quad \{y = \max(0, 2\bar{i})\}} \quad (2)$$

$$D_1 = \frac{\vdash A \Rightarrow A' \quad \frac{}{\{A'\} \quad y := -x \quad \{C\}} \quad (4) \quad \vdash C \Rightarrow C}{\{x = \bar{i} \wedge x \leq 0\} \quad y := -x \quad \{C\}} \quad (6)$$

$$D_2 = \frac{\vdash B \Rightarrow B' \quad \frac{}{\{B'\} \quad y := x \quad \{C\}} \quad (4) \quad \vdash C \Rightarrow C}{\{x = \bar{i} \wedge \neg(x \leq 0)\} \quad y := x \quad \{C\}} \quad (6)$$

$c_3 = (\text{if } x \leq 0 \text{ then } y := -x \text{ else } y := x)$

$C \equiv y = \max(0, 2\bar{i})[y + x/y]$

$A \equiv x = \bar{i} \wedge x \leq 0$

$A' \equiv C[-x/y]$

$B \equiv x = \bar{i} \wedge \neg(x \leq 0)$

$B' \equiv C[x/y]$

2)

$$\frac{\frac{\vdash I \wedge b \Rightarrow I' \quad \frac{}{\{I'\} \quad x := x + 1 \quad \{C\}} \quad (4) \quad \vdash C \Rightarrow C}{\{I \wedge b\} \quad x := x + 1 \quad \{C\}} \quad (6) \quad \frac{}{\{C\} \quad y := y - 1 \quad \{I\}} \quad (4)}{\frac{\{I \wedge b\} \quad (x := x + 1; y := y - 1) \quad \{I\}}{\{I\} \quad c_2 \quad \{\neg b \wedge I\}} \quad (5)}{\frac{\vdash A \Rightarrow I \quad \frac{}{\{A\} \quad c_2 \quad \{B\}} \quad (6) \quad \vdash \neg b \wedge I \Rightarrow B}{\vdash A \Rightarrow B}} \quad (6)$$

$$\begin{aligned}
c_2 &= (\text{while } x \leq y - 1 \text{ do } (x := x + 1; y := y - 1)) \\
b &= (x \leq y - 1) \\
A &\equiv (x = 0) \wedge (y = \bar{n}) \wedge (\bar{n} \geq 0) \\
B &\equiv y = \lfloor \bar{n}/2 \rfloor \\
C &\equiv I[y - 1/y] \\
I &\equiv (x + y = \bar{n}) \wedge (y \geq \lfloor \bar{n}/2 \rfloor) \\
I' &\equiv C[x + 1/x] \\
&\equiv (x + 1 + y - 1 = \bar{n}) \wedge (y - 1 \geq \lfloor \bar{n}/2 \rfloor)
\end{aligned}$$

3) (a)

$$\begin{aligned}
A &\equiv (y = \bar{n}) \wedge (\bar{n} > 0) \wedge (z = 0) \\
B &\equiv z = \lfloor \log_2 \bar{n} \rfloor \\
I &\equiv z = \lfloor \log_2 \bar{n} \rfloor - \lfloor \log_2 y \rfloor \wedge y > 0 \\
I' &\equiv z + 1 = \lfloor \log_2 \bar{n} \rfloor - \lfloor \log_2 \lfloor \frac{y}{2} \rfloor \rfloor \wedge \lfloor \frac{y}{2} \rfloor > 0 \\
b &= 2 \leq y \\
\neg b &= 2 > y
\end{aligned}$$

(b)

$$\begin{aligned}
A &\equiv (y = \bar{n}) \wedge (\bar{n} \geq 0) \wedge (t = 1) \wedge (z = 0) \\
B &\equiv z = \bar{n} \\
I &\equiv (y \cdot t + z = \bar{n}) \wedge y \geq 0 \\
I' &\equiv (\lfloor \frac{y}{2} \rfloor \cdot 2t + z + t(y \% 2) = \bar{n}) \wedge \lfloor \frac{y}{2} \rfloor \geq 0 \\
b &= 1 \leq y \\
\neg b &= 1 > y
\end{aligned}$$